Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3391-WE01

Title: (BEng/MEng)

Control and Signal Processing 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown. Use the graph paper provided for Question 3 (2) (c) and Question 4 (2) (c) in Section B. Place the graph paper with your answer to the questioninside your examination answer book making sure to write your anonymised Examination Candidate Code clearly at the top of the page and use the treasury tag provided to secure your graph paper inside the exam answer booklet.

Revision:	

Section A: Question 1

(1) Define and differentiate between the following types of signals. Use relevant illustrations and examples to support your answers.

(a) Deterministic and Random time signals

(b) Continuous and Discrete time signals

(c) Periodic and non-Periodic time signals

[28%]

(2) Consider a linear time-invariant (LTI) system.

(a) Define the two core properties of an LTI system: linearity and time-invariance.

(b) What is the relationship between the unit step u(t) and the impulse $\delta(t)$ functions?

[20%]

(3) Consider an LTI system illustrated in Fig.Q1.1 where the input and output discrete time signals are $x_1[n]$ and $x_2[n]$ and $y_1[n]$ and $y_2[n]$, respectively. The analytical expression of the signals $x_1[n]$ and $x_2[n]$ is given by equations (eq1.1) and (eq1.2), respectively.

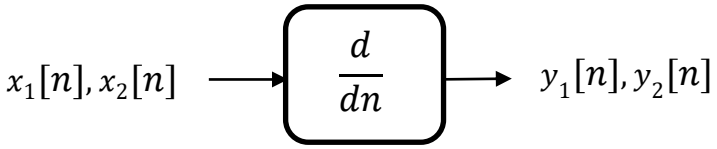


Fig.Q1.1

$$x_1[n] = 2u[n-1] + 5u[n-3] - 3u[n-4] - u[n-5] \quad \text{(eq1.1)}$$

$$x_2[n] = 3u[n] + 2u[n-2] - u[n-3] \quad \text{(eq1.2)}$$

(a) Using the relation between the unit step and the impulse functions in an LTI system, determine the expression of the discrete output signals $y_1[n]$] and $y_2[n]$ as a function of the impulse function, respectively.

(b) Use the results from Q1.3(a) to extract the discrete numerical values of $y_1[n]$ and $y_2[n]$ in the form of a matrix, e.g. {a b c ...}. Sketch the discrete output signals $y_1[n]$ and $y_2[n]$.

(c) Compute the convolution z[n] of these discrete output signals: $z[n] = y_1[n] * y_2[n]$.

[52%]

continued

Section A: Question 2

(1) Consider $x(t) \leftrightarrow \mathrm{X}(\omega)$ *and* $x(-t) \leftrightarrow \mathrm{X}^*(\omega)$. Apply the basic operators' properties of an LTI system to determine the Fourier transforms of the following signals:

(a) $x_1 = x(T - t)$

(b) $x_2 = x(2t - T)$

(c) $x_3 = \frac{d}{dx}[x_1(t)^* x_2(t)]$

[52%]

(2) A radio station operates in amplitude modulation (AM) using the signals $x(t)$ and $y(t)$ in an electronic mixer as illustrated in Fig.Q2.1. The modulation frequency is f_m = 5 kHz and the carrier frequency is f_c = 0.5 MHz. $x(t) = \mathrm{k}[1 + mcos(\omega_m t)]$ and $y(t)$ = cos$(\omega_c t)$.

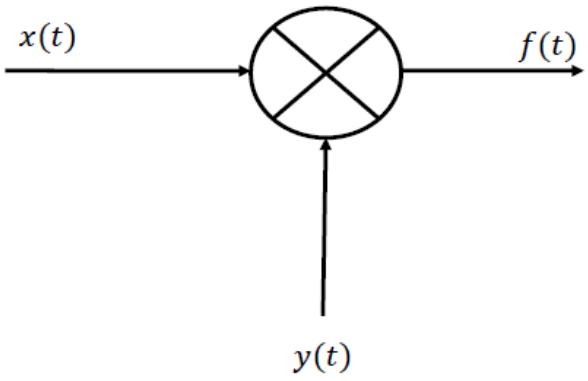


Fig.Q2.1: Electronic signals mixer

(a) Determine the explicit analytical expression of the output signal $f(t)$.

(b) Assuming m < 1 sketch both the single-sided spectrum (SSS) and the double-sided spectrum (DSS) of the full AM signal.

(c) Comment on the values of the frequencies and amplitudes represented in the SSS and DSS spectra and on the correlation between SSS and DSS.

[48%]

continued

Section B: Question 3

(1) A system is shown in Figure Q3.1 below. State whether this system is stable and give your reason.

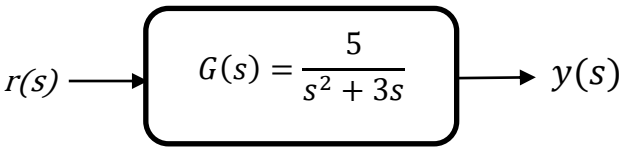


Figure Q3.1 A system with transfer function G(s)

[10%]

(2) Adding a negative feedback to this system will give a closed-loop system as in Figure Q3.2.

(a) Give the closed-loop transfer function, the closed-loop poles, the damping ratio, and the undamped natural frequency for the closed-loop system in Figure Q3.2.

[20%]

(b) Compare the closed-loop system and the open-loop system in terms of performance, complexity and cost.

[10%]

(c) Use the provided graph paper to sketch the root locus of the closed-loop system in Figure Q3.2, showing all necessary steps of the plotting.

[10%]

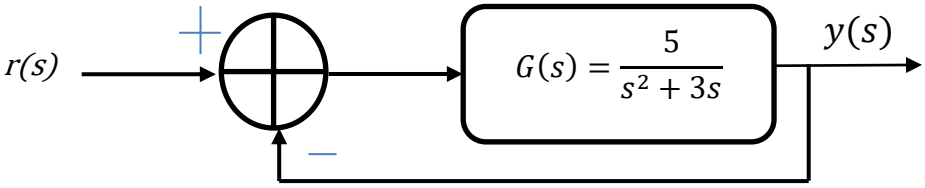


Figure Q3.2 A closed-loop with negative unit feedback

(3) A practical phase-lead compensator $G_c(s) = K_c \frac{s+z_c}{s+p_c}$ is added to the system in Figure Q3.2 to become a controlled closed-loop system as in Figure Q3.3,

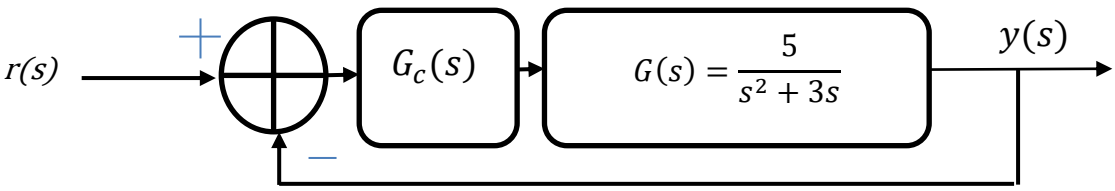


Figure Q3.3 A closed-loop control system with negative unit feedback

continued

(a) Design this phase-lead compensator by determining the values of K_c, z_c, p_c to meet the requirements of both the overshoot percentage to its final value of less than 10% and the settling time of 4% error to its final value in less than 2 seconds.

[30%]

(b) Compare the forward difference, backward difference and Tustin's rule in terms of stability of the mapped system. Discretise the designed compensator $G_c(s)$ in the previous part by using the Tustin's rule with $z = \frac{1+sT/2}{1-sT/2}$, where $T = 0.4$ seconds is the sampling interval, to a form suitable for implementation on a microcontroller.

[20%]

Section B: Question 4

(1) The stability of a linear time-invariant system can be analysed in the time domain, the s-plane and the frequency domain. State the advantages of using the frequency domain approach over the time domain and s-plane approaches in the determination of system stability.

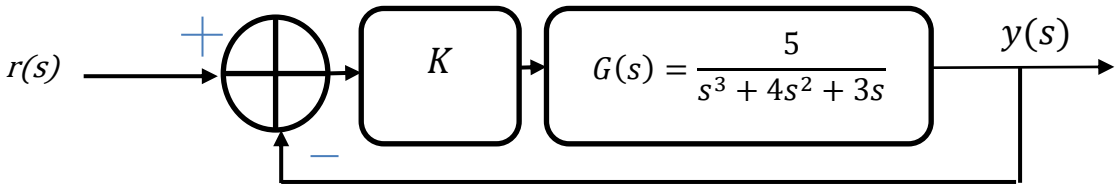


Figure Q4 A closed-loop control system

[20%]

(2) A closed-loop control system is given in Figure Q4 above, where $K = 1$.

(a) Discuss the three methods of frequency domain approaches, Nyquist diagram, Bode diagram and Nichols chart, in terms of their features, advantages and disadvantages.

[20%]

(b) Calculate the open-loop frequency response $G(j\omega)$ in the complex number form of $a(\omega) + jb(\omega)$ with real part $a(\omega)$ and imaginary part $b(\omega)$. Also, calculate the gain in dB and the phase $\varphi(\omega)$ of the open-loop frequency response $G(j\omega)$.

[20%]

(c) Using the expressions obtained in Part (b) and the key values of $\omega = 0, 0.1, 1, 2, 10$, plot the Nyquist diagram and the Bode diagram of the open-loop frequency response $G(j\omega)$.

[20%]

(d) Using either your Nyquist diagram or Bode diagram obtained in Part (c), find the gain margin, the phase margin, the phase cross-over frequency and the gain cross-over frequency. (Note that you only need to use one of the diagrams, not both).

[20%]

continued

Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) \;=\; a_o \;+\; \sum_{k=1}^{\infty} [a_k \cos(k\,\omega_o t) \;+\; b_k \sin(k\,\omega_o t)]$$

where $\omega_0 \;=\; \frac{2\pi}{T} = 2\pi f_0$

$$a_o \;=\; \frac{1}{T} \int_T x(t)dt \;=\; dc\ term$$

$$a_k \;=\; \frac{2}{T} \int_T x(t) \cos(k\omega_o t)\, dt$$

$$b_k \;=\; \frac{2}{T} \int_T x(t) \sin(k\omega_o t)\, dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) \;=\; a_o \;+\; \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2} \;\cos(k\omega_o t \;+\; \phi_k)$$

where $\varphi_k \;=\; -\,tan^{-1}\left(\frac{b_k}{a_k}\right)$

Complex exponential form

$$x(t) \;=\; \sum_{k=-\infty}^{\infty} d_k \; e^{j(k\omega_o)t}$$

continued

Where $d_k = \frac{1}{T} \int_T x(t)e^{-j(k\omega_o)t}\, dt$

and $d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

Fourier transform

$$F\{x(t)\} = X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt$$

Alternatively, it can be written in terms of the frequency in cycles per seconds or Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\, dt$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$

or

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft} df$$

Fourier transform properties

1. Linearity

   $x_1(t) \leftrightarrow X_1(\omega)$ and $x_2(t) \leftrightarrow X_2(\omega)$ then

   $ax_1(t) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$

2. Differentiation

   $$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. Integration

   $$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega} X(\omega) + \pi X(0)\delta(\omega)$$

4. Time scaling

   $$x(bt) \leftrightarrow \frac{1}{|b|} . X\left(\frac{\omega}{b}\right)$$

continued

5. Time reversal
   $x(-t) \;\leftrightarrow\; X(-\omega) = X^*(\omega)$

6. Time delay
   $\mathrm{y}(t) = x(t-T) \leftrightarrow X(\omega)\, e^{-\mathrm{j}\omega T}$

7. Multiplication in time
   $x_1(t).x_2(t) \;\leftrightarrow\; \frac{1}{2\pi}\left(X_1(\omega) * X_2(\omega)\right)$

8. Multiplication in frequency (convolution in time)
   $x_1(t) * x_2(t) \leftrightarrow X_1(\omega).\, X_2(\omega)$

9. Modulation property
   $x(\mathrm{t})\cos(\omega_0 t) \leftrightarrow \frac{1}{2}X(\omega-\omega_0) + \frac{1}{2}X(\omega+\omega_0)$

10. Duality property

    If $x(t) \leftrightarrow X(\omega)$

    Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series (DFS)

$$d_k = X(k) = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{\frac{-j2\pi kn}{N}}$$

Inverse Discrete Fourier Series (IDFS)

$$X(n) = \sum_{n=0}^{N-1} d_k\, e^{\frac{+j2\pi kn}{N}}$$

Euler's identities

$$A.sin(\omega_o t) = A\left(\frac{e^{j\omega_o t} - e^{-j\omega_o t}}{2j}\right)$$

$$A.cos(\omega_o t) = A\left(\frac{e^{j\omega_o t} + e^{-j\omega_o t}}{2}\right)$$

$$A.cos(\omega_o t + \phi_0) = A\left(\frac{e^{j\omega_o t}e^{j\phi_0} + e^{-j\omega_o t}e^{-j\phi_0}}{2}\right)$$

continued

$e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k].x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k].x_2[k]$$

Convolution with an impulse

$$x(t) * \delta(t-t_o) = x(t-t_o)$$

$$y(t) = x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = x(t)\int_{-\infty}^{\infty} \delta(t-\tau)d\tau = x(t)$$

continued

Correlation functions

Continuous time auto-correlation

Power signal:

$$P \propto lim \frac{1}{2T} \int_{-\infty}^{\infty} x(t)x(t+\tau)d\tau$$

Energy signal:

$$E = \int_{-\infty}^{\infty} x(t)x(t+\tau)d\tau$$

Fourier transform of an impulse

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$$\delta(\omega) \leftrightarrow \frac{1}{2\pi}$$

Roots of a second order polynomial

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

$$\cos(\theta) = \cos(-\theta) = \sin\left(\theta - \frac{\pi}{2}\right)$$

$$\sin(\theta) = -\sin(-\theta) = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\tan(\theta) = -\tan(-\theta) = \cot\left(\frac{\pi}{2} - \theta\right)$$

$$\sin(2\theta) = 2\ \cos(\theta)\sin(\theta)$$

$$\cos(2\theta) = cos^2(\theta) - \ sin^2(\theta) = 2\,cos^2(\theta) - \ 1 = 1 - 2\,sin^2(\theta)$$

continued

$$\tan(2\theta) = \frac{2\tan(\theta)}{1 - tan^2(\theta)}$$

$$\sin\alpha.\sin\beta = \frac{1}{2}\{\cos(\alpha - \beta) - cos(\alpha + \beta)\}$$

$$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha - \beta) + cos(\alpha + \beta)\}$$

$$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha - \beta) + sin(\alpha + \beta)\}$$

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - \cos(\theta)}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 + \cos(\theta)}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin(\theta)}{1 + \cos(\theta)}$$

$$cos^2(\theta) + sin^2(\theta) = 1$$

$$sec^2(\theta) - stan^2(\theta) = 1$$

$$cosec^2(\theta) - cot^2(\theta) = 1$$

$$sin^2(\theta) = \frac{1}{2}(1 - \cos(2\theta)$$

$$cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta)$$

$\tan^2\theta = (1 - \cos 2\theta)/(1 + \cos 2\theta)$

$$tan^2(\theta) = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

$$\sin(A + B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\sin(A - B) = \sin(A)\cos(B) - \cos(A)\sin(B)$$

$$\cos(A + B) = \cos(A)\cos(B) - \sin(A)\sin(B)$$

$$\cos(A - B) = \cos(A)\cos(B) + \sin(A)\sin(B)$$

$$\tan(A + B) = \frac{\tan(A) + \tan(B)}{1 - \tan(A)\tan(B)}$$

$$\tan(A - B) = \frac{\tan(A) - \tan(B)}{1 + \tan(A)\tan(B)}$$

continued

$$\sin(A)+\sin(B)=2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin(A)-\sin(B)=2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\cos(A)+\cos(B)=2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\cos(A)-\cos(B)=-2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\tan(A)+\tan(B)=\frac{\sin(A+B)}{\cos(A)\cos(B)}$$

$$\tan(A)-\tan(B)=\frac{\sin(A-B)}{\cos(A)\cos(B)}$$

$$\sin(2A)+\sin(2B)=1-\cos(A+B)\cos(A-B)$$

$$\sin(2A)-\sin(2B)=\sin(A+B)\sin(A-B)$$

$$\cos(2A)+\sin(2B)=1-\sin(A+B)\sin(A-B)$$

$$\cos(2A)-\sin(2B)=\cos(A+B)\cos(A-B)$$

$$\cos(2A)+\cos(2B)=1+\cos(A+B)\cos(A-B)$$

$$\cos(2A)-\cos(2B)=-\sin(A+B)\sin(A-B)$$

Triangle

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

$$a^2=b^2+c^2-2bc\cos(A)$$

$$\frac{a}{\sin(A)}=\frac{b}{\sin(B)}=\frac{c}{\sin(C)}$$

$$\frac{a-b}{a+b}=\frac{\tan\left(\frac{A-B}{2}\right)}{\tan\left(\frac{A+B}{2}\right)}$$

continued

Integration by parts

$$\int u(t).v(t)dt = u(t)\int v(t)\,dt - \int u'(t).d(v(t))dt$$

$$u(t) = u \;\; and \;\; v(t) = v$$

$$\int u(t).v(t)dt = u\int vdt - \int u'.\int d(v)\,dt + c$$

or

$$\int u.dv = uv - \int v\,du$$

Integration of trigonometric & Exponential functions

$$\int \sin(at)dt = -\frac{1}{a}.\cos(at) + c$$

$$\int \cos(at)dt = \frac{1}{a}.\sin(at) + c$$

$$\int \exp(at)dt = \frac{1}{a}.\exp(at)$$

Derivation of trigonometric & Exponential functions

$$\frac{d\{\sin(at)\}}{dt} = a.\cos(at)$$

$$\frac{d\{\cos(at)\}}{dt} = -a.\sin(at)$$

$$\frac{d\{\exp(at)\}}{dt} = a.\exp(at)$$

continued

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

7 Steps for Root Locus Sketching:

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

continued

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle Φ

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\varphi = \frac{(2q + 1)}{n - m}\pi$$

where $q = 0,1, \dots, (n - m - 1)$.

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n} |s - p_i|}{\prod_{j=1}^{m} |s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n} \phi_{p_i} - \sum_{j=1}^{m} \phi_{z_j} = (2q + 1)\pi$$

END